

MDM Architecture and Operations

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MDM overview

MDM receives meter information from upstream systems and provides controlled storage and processing.

Data ingestion

Data may include interval data, register reads, events, and approved meter datasets. Interfaces should be monitored for delays and failures.

Validation

Validation can identify missing intervals, duplicate records, abnormal consumption, timestamp problems, and invalid values.

Estimation

Where approved, missing or invalid data may be estimated or edited. Estimated values must remain distinguishable from actual readings.

Storage

MDM stores processed data used by billing, analytics, reporting, and operations. Data quality and lineage should be retained where required.

Downstream use

Downstream users should consider data-quality status before using readings for billing or analysis.

Operational note

This is synthetic content created for local RAG development, testing, interview demonstration, and GitHub publication. It is not a confidential utility procedure or contractual document.